📘 AWS EBS Volume – Overview and How to Increase Size

1️⃣ What is an EBS Volume?

Amazon EBS (Elastic Block Store) is a **block storage service** in AWS used with EC2 instances.

- It works like a **virtual hard disk** attached to a server.
- It stores:
  - Operating System files
  - Application data
  - Logs
  - Databases

Example Structure:

```
EC2 Instance
   |
   └── EBS Volume (Storage Disk)
```

2️⃣ Why Increase EBS Volume Size?

Sometimes disk space becomes full.

Common reasons:

- Application logs increase over time
- Database data grows
- Docker images and containers use space
- Application files and uploads increase

Example:

```
Total Disk : 20GB
Used Disk  : 19GB
Free Space : 1GB
```

3️⃣ Troubleshooting (Before Increasing EBS)

Before increasing the volume, check where the disk space is used.

Check Disk Space:

df -h

Shows:

- Total size
- Used space
- Available space

Example:

Filesystem   Size Used Avail Use%
/dev/nvme0n1p1   20G   19G   1G 95%

Check Folder Size:

du -sh /*

Example:

2.1G   /var
5.3G   /home
7.2G   /opt

Check Logs:

du -sh /var/log

👉 Delete unnecessary files if possible.

4️⃣ Steps to Increase EBS Volume Size

Step 1 – Open EC2

- Login to AWS Console
- Go to EC2 Dashboard
- Click **Instances**
- Select your instance

Step 2 – Open Storage

- Scroll down
- Click **Storage tab**
- Find attached **Volume ID**
- Click on Volume ID

Example:

Volume ID: vol-xxxxxxxx

Step 3 – Modify Volume

- Click **Modify**
- Enter new size

Example:

Old Size : 20 GB
New Size : 40 GB

- Click **Modify** and confirm

👉 AWS will start resizing:

modifying → optimizing → completed

5️⃣ Extend File System on Server

After increasing size in AWS, update the filesystem in Linux.

Step 1 – Connect to Server

ssh -i key.pem ubuntu@server-ip

Step 2 – Check Disk Info

Lsblk
df -h

Step 3 – Extend Partition

sudo growpart /dev/nvme0n1 1

This command expands (increases) the size of a partition after you increase disk size in AWS.

Step 4 – Resize Filesystem

sudo resize2fs /dev/nvme0n1p1

this command resizes (expands or shrinks) the filesystem on the partition `/dev/nvme0n1p1` so it can use the full space of the disk.

Step 5 – Verify

df -h

6️⃣ Important Commands

`lsblk`

- command is used to list all block storage devices like disks and partitions in a tree format.

`df -h`

- is used to check **disk space usage** in the system.

`du -sh`

- is used to show total size of a directory in human-readable format.

7️⃣ Disk Example (lsblk Output)

```
nvme0n1      8G  disk
├─nvme0n1p1  7G  part /
nvme1n1      8G  disk /data
```

Explanation:

Name	Description
nvme0n1	First disk (root EBS volume)
nvme0n1p1	Partition of disk
nvme1n1	Second attached EBS volume

✅ Final Summary

- EBS is a **virtual disk in AWS**
- Increase size when disk is full
- Always check usage before increasing
- After resizing in AWS, update filesystem in Linux

👉 **Process:**

Check → Increase Volume → Extend Partition → Resize Filesystem → Verify

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